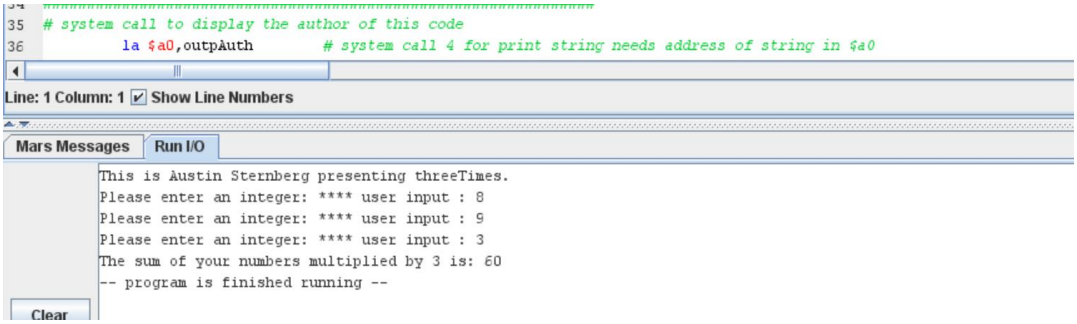
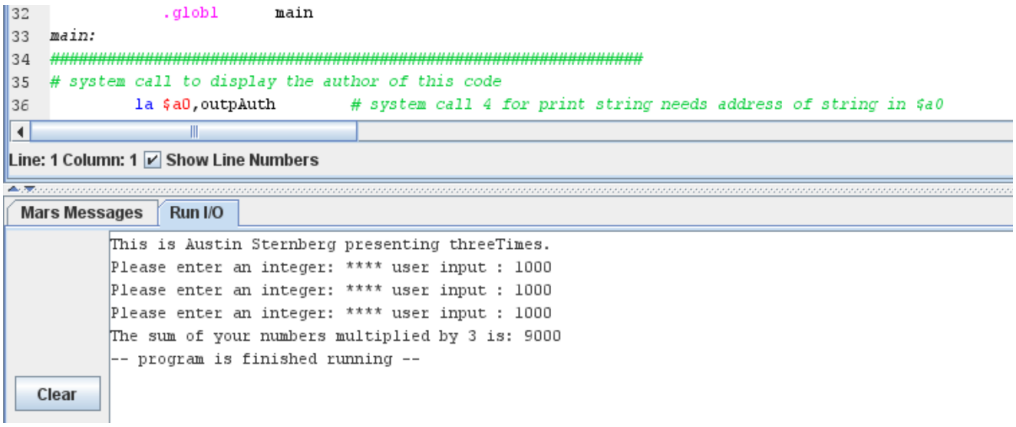
Project 1 CO-1 Report – Three Times

1. Austin Sternberg
2. Code File: threeTimes-1.asm
3. I choose to take a linear approach to this problem. Building off the starter code I was given, I created a couple lines that prompt for user input and then print that input to the console using system call 5 and a register to hold it. I then repeated that, along with the original system call 4 to give the user instructions, two more times for the other two integers, choosing the next register for each subsequent integer. From there it was simple as adding the integers together, then multiplying that sum by 3. To do so, I stored the result of the first integer and second integer (stored in $t1 & $t2 respectively) to another register ($t4). From here I needed to add the value to $t4 to the final integer ($t3), and store that value back into $t4. For the multiplication part, I choose to store the constant (3) in another register, $t5, then I read ahead to implement the mul function to multiply the contents of $t4 and $t5, storing that value in $t6. From here it was as simple as moving the value of $t6 to $a0 for output and using system call 1 to print the integer after the output of the final text line.

4.





5. With this project I needed to dive into programming in MIPS for the first time. Now I felt reasonably confident given what we had studied in class, but this project had some challenges. Mainly this was around the system calls required to print different types of data and to prompt for user input. This turned out to not be a huge problem, as we had been given some reference material in class that I was able to read and understand what needed to be done. The other difficult part was learning how to use the ‘mul’ function, though similarly I was able to do a bit of research on the MIPs Language and figure out how this function was supposed to be called.